

2c. Content of Physical systems (H481/01)

The Physical systems (01) component is built around two main topics, Landscape Systems and Earth's Life Support Systems. Learners will explore **one** chosen landscape from three options, coastal landscapes, glaciated landscapes and dryland landscapes, as well as the carbon and water cycles in a systems framework.

The inter-relationships between the land, oceans and atmosphere help learners to understand the processes, characteristics and impacts on these

landscapes and cycles, which shape them over time and create a number of issues when attempting to manage them.

Learners will investigate examples and case studies at a range of scales to understand the dynamic nature of landscapes and the water and carbon cycles. Quantitative and qualitative fieldwork opportunities present themselves within this component, within helping learners to generate new knowledge and understanding about the real world.

Topic 1.1 – Landscape Systems

“Life is like a landscape. You live in the midst of it but can describe it only from the vantage point of distance.” – Charles Lindbergh

This topic introduces learners to the integrated study of Earth surface processes, landforms and resultant landscapes within the conceptual framework of a systems approach.

An understanding of Earth surface processes, together with their associated transfers of energy and movements of materials underpins the landscape systems topic.

Choosing **one** of either **Option A** coastal, **Option B** glaciated or **Option C** dryland landscapes, learners will explore how the landscape can be viewed as system, how landforms developed within their chosen landscape and the influences of both climate and human activity on this.

For all options within this topic, study **must** include at least **two** case studies from landscapes beyond the UK and at least **one** landscape from the UK.

1.1.1 Option A – Coastal Landscapes

1. How can coastal landscapes be viewed as systems?	
Key Ideas	Content
1.a. Coastal landscapes can be viewed as systems.	• A conceptual overview of: ◦ the components of coastal landscape systems, including inputs, processes and outputs ◦ the flows of energy and material through coastal systems ◦ sediment cells.
1.b. Coastal landscape systems are influenced by a range of physical factors.	• Potential influences on coastal landscape systems of: ◦ winds, including speed, direction and frequency ◦ waves, including wave formation, development and breaking ◦ tides, including tidal cycles and range ◦ geology, including lithology and structure ◦ global pattern of ocean currents.
1.c. Coastal sediment is supplied from a variety of sources.	• The various sources of coastal sediment: ◦ terrestrial, including fluvial deposition, weathering and mass movement, marine erosion, aeolian deposition and longshore drift ◦ offshore, including marine deposition ◦ human, including beach nourishment.

2. How are coastal landforms developed?

Key Ideas	Content
2.a. Coastal landforms develop due to a variety of interconnected climatic and geomorphic processes.	• The influence of flows of energy and materials on geomorphic processes, including weathering, mass movement, wave, fluvial and aeolian erosion, transportation and deposition. • The formation of distinctive landforms, predominantly influenced by erosion, including bays, headlands, cliffs, shore platforms, geos, blow holes, caves, arches, stacks and stumps. • The formation of distinctive landforms, predominantly influenced by deposition, including beaches, spits, on-shore bars, tombolos and salt marshes.
2.b. Coastal landforms are inter-related and together make up characteristic landscapes.	• Case studies of one high energy coastline (such as rocky) and one low energy coastline, such as estuarine, to illustrate: ◦ the physical factors which influence the formation of landforms within the landscape system ◦ the inter-relationship of a range of landforms within the characteristic landscape system ◦ how and why the landscape system changes over time from millennia to seconds, such as cliff collapse in seconds, seasonal changes in beach profile and spit growth over millennia. At least one of the case studies must be from beyond the UK.

3. How do coastal landforms evolve over time as climate changes?

Key Ideas	Content
3.a. Emergent coastal landscapes form as sea level falls.	• How landforms in emergent landscapes are influenced by falling sea levels due to a cooling climate, including: ◦ climate changes that occurred during a previous time period and the resultant sea level fall ◦ the influence of sea level fall and geomorphic processes in shaping landforms, including raised beaches, marine terraces and abandoned cliffs ◦ the modification of these landforms by processes associated with present and future climate and sea level changes.
3.b. Submergent coastal landscapes form as sea level rises.	• How landforms in submergent landscapes are influenced by rising sea level due to a warming climate, including: ◦ climate changes that occurred during a previous time period and the resultant sea level rise ◦ the influence of sea level rise and geomorphic processes in shaping landforms, including rias, fjords and shingle beaches ◦ the modification of these landforms by processes associated with present and future climate and sea level changes.

4. How does human activity cause change within coastal landscape systems?

Key Ideas	Content
4.a. Human activity intentionally causes change within coastal landscape systems.	• Case study of one coastal landscape that is being managed, including: ◦ the management strategy being implemented and the reason for its implementation, such as groyne construction or off-shore dredging ◦ their intentional impacts on processes and flows of material, processes and/or energy through the coastal system, such as their effect on the sediment budget ◦ the effect of these impacts in changing coastal landforms, such as changes in beach profile ◦ the consequence of these changes on the landscape, such as extension of the coastal landscape seawards.
4.b. Economic development unintentionally causes change within coastal landscape systems.	• Case study of one coastal landscape that is being used by people to illustrate: ◦ the economic development taking place and the reasons for it taking place, such as trade routes, port or tourist resort development ◦ their unintentional impacts on processes and flows of material, processes and/or energy through the coastal system, such as disturbance to the sediment cell balance ◦ the effect of these impacts in changing coastal landforms, such as beach profiles ◦ the consequence of these changes on the landscape, such as coastal retreat or protection.

5. Topic-specific skills:

- observation skills
- measurement and geo-spatial mapping skills
- data manipulation and statistical skills applied to field measurements
- sediment budget calculations
- mass balance calculations.